① $x^2 - 3x + 1 = f(x)$

a) $x^2 - 3x + 1 = 0$

$x^2 - 2x + 1 = x \rightarrow x_{k+1} = x_k^2 - 2x_k + 1$

$x_0 = 1$

$x_1 = \cancel{0^2 - 2 \cdot 0 + 1 =} \; 1^2 - 2 \cdot 1 + 1 = 0$

$x_2 = 0^2 - 2 \cdot 0 + 1 = 1$

$\varphi'(x) = 2x - 2$

$|\varphi'(1)| = 2 - 2 = 0 \rightarrow$ ~~[illegible]~~ konverguje

b)

~~[illegible]~~

$x = \sqrt{3x+1}$

$x_{k+1} = \sqrt{3x_k + 1}$

$x_0 = 1$

$x_1 = \sqrt{3+1} = 2$

$x_2 = \sqrt{6+1} = \sqrt{7}$

$\varphi'(x) = \dfrac{3}{2\sqrt{3x+1}}$

$|\varphi'(1)| = \dfrac{3}{2\sqrt{3+1}} = \dfrac{3}{4} < 1 \rightarrow$ konverguje